Data Dictionary — Coffee Bilateral Trade (BACI/CEPII)

File: `coffee_bilateral_trade_BACI.csv`
Source: BACI — Base pour l'Analyse du Commerce International (CEPII)
Version: 202601 (released January 22, 2026)
HS Revision: HS92
License: Etalab Open Licence 2.0 — free to use with source attribution
Reference: Gaulier, G. and Zignago, S. (2010). *BACI: International Trade Database at the Product-Level.* CEPII Working Paper N°2010-23.

What's Included

Dimension	Coverage
Years	1995 – 2024 (30 years)
Total rows	136,768 bilateral trade flows
Unique exporters	229 countries
Unique importers	233 countries
HS codes	090111 (green/unroasted, non-decaffeinated) · 090112 (unroasted, decaffeinated)
File size	~15.7 MB

Row counts by HS code:

HS Code	Description	Rows
090111	Coffee: not roasted or decaffeinated (standard green coffee)	103,734
090112	Coffee: decaffeinated, not roasted	33,034

Row counts by year grow steadily from 2,600 (1995) to a peak of 6,790 (2023), reflecting expanding global trade reporting coverage over time.

Column Definitions

Year

Property	Detail
Type	Integer
Nullable	No

Property	Detail
Range	1995 – 2024
Unique values	30
Description	Calendar year of the trade flow. All years follow the standard January–December reporting period. Note that 2024 data may be more preliminary than earlier years, as BACI is updated annually each January using the prior year's Comtrade submissions.

`Exporter_Code`

Property	Detail
Type	Integer
Nullable	No
Unique values	229
Description	UN Comtrade 3-digit numeric country code for the exporting country. Follows the UN M49 standard but may differ from ISO 3166-1 numeric codes in some cases. Use `Exporter_ISO3` for standard cross-referencing with other datasets.

`Exporter_Name`

Property	Detail
Type	String (text)
Nullable	No
Unique values	229
Description	Full English name of the exporting country, sourced from the BACI country metadata file (`country_codes_V202601.csv`). Names follow UN Comtrade conventions and may differ slightly from ISO or World Bank naming (e.g., "Viet Nam" rather than "Vietnam").
Examples	Brazil, Viet Nam, Colombia, Honduras, Ethiopia, Indonesia, Guatemala, Uganda, Peru

`Exporter_ISO3`

Property	Detail
Type	String (3 characters)
Nullable	No (may be empty string for a small number of non-standard territories)
Unique values	229

Property	Detail
Description	ISO 3166-1 alpha-3 country code for the exporter. Useful for joining with other datasets (World Bank, FAO, geospatial shapefiles).
Examples	BRA, VNM, COL, HND, ETH, IDN, GTM, UGA, PER

`Importer_Code`

Property	Detail
Type	Integer
Nullable	No
Unique values	233
Description	UN Comtrade 3-digit numeric country code for the importing country. Same coding standard as `Exporter_Code`.

`Importer_Name`

Property	Detail
Type	String (text)
Nullable	No
Unique values	233
Description	Full English name of the importing country. Slightly more unique values than exporters (233 vs 229) because some territories appear only as importers in the coffee trade.
Examples	USA, Germany, Italy, Japan, Switzerland, Belgium, Spain, France, Netherlands, Canada

`Importer_ISO3`

Property	Detail
Type	String (3 characters)
Nullable	No
Unique values	233
Description	ISO 3166-1 alpha-3 country code for the importer. Use for geospatial joins and cross-dataset merges.
Examples	USA, DEU, ITA, JPN, CHE, BEL, ESP, FRA, NLD, CAN

`HS_Code`

Property	Detail
Type	String (6 characters)
Nullable	No
Unique values	2
Description	6-digit Harmonized System product code under the HS92 revision. Must be treated as a string (not a number) to preserve the leading zero. The two codes cover the entirety of unroasted coffee bean trade — roasted, instant, and other coffee products are excluded from this dataset.

Allowed values:

Code	Full HS Description
090111	Coffee: not roasted or decaffeinated — the dominant global green coffee commodity
090112	Coffee: decaffeinated, not roasted — specialty/health segment, smaller volume

`HS_Description`

Property	Detail
Type	String (text)
Nullable	No
Unique values	2
Description	Human-readable description of the HS product code, sourced from the BACI product metadata file. Provided for convenience; `HS_Code` is the authoritative identifier.

`Value_1000USD`

Property	Detail
Type	Float
Nullable	No
Non-null	136,768 (100%)
Minimum	0.001
Maximum	1,908,635.123
Mean	3,793.53
Median	36.30

Property	Detail
Description	Value of the bilateral trade flow expressed in thousands of current US dollars (nominal prices). To convert to full USD, multiply by 1,000. Values are not inflation-adjusted. The distribution is extremely right-skewed — the median flow is ~$36,300 while the mean is ~$3.8M, driven by a small number of very large country-pair flows. The maximum recorded single flow (~$1.9B) represents Brazil's coffee exports to Germany in a peak year.

`Quantity_tonnes`

Property	Detail
Type	Float
Nullable	Yes
Non-null	135,777 (99.3%)
Null	991 (0.7%)
Minimum	0.001
Maximum	456,542.494
Mean	1,398.95
Median	11.29
Description	Physical weight of the trade flow in metric tonnes. Missing in approximately 0.7% of rows — this is a known characteristic of BACI documented by CEPII, typically occurring when the reporting country submitted value data but not weight. When both `Value_1000USD` and `Quantity_tonnes` are available, their ratio yields the unit value (price per tonne), a useful proxy for coffee quality or price tier.

Top Bilateral Flows — 2023

Ranked by `Value_1000USD`, filtered to year 2023:

Rank	Exporter	Importer	HS Code	Value (1000 USD)	Quantity (tonnes)
1	Brazil	USA	090111	1,222,301	324,479
2	Colombia	USA	090111	1,155,000	234,560
3	Brazil	Germany	090111	1,122,402	313,417
4	Brazil	Italy	090111	773,759	193,419
5	Brazil	Japan	090111	455,032	122,858
6	Viet Nam	Germany	090111	454,576	193,031
7	Honduras	USA	090111	443,051	100,569
8	Guatemala	USA	090111	426,045	74,815
9	Viet Nam	Italy	090111	370,689	148,502

Rank	Exporter	Importer	HS Code	Value (1000 USD)	Quantity (tonnes)
10	Viet Nam	USA	090111	331,715	127,976

Top exporters by total value (2023):

Exporter	Total Value (1000 USD)
Brazil	7,936,945
Viet Nam	3,548,507
Colombia	3,083,467
Honduras	1,524,942
Ethiopia	1,299,730
Indonesia	1,088,296
Guatemala	1,066,713
Uganda	995,450
Peru	886,486
Germany	799,916

> Note: Germany appears as a top exporter because it re-exports processed/traded coffee — it is also the #2 importer.

Top importers by total value (2023):

Importer	Total Value (1000 USD)
USA	5,719,315
Germany	3,070,979
Italy	2,143,602
Japan	1,405,268
Switzerland	1,052,916
Belgium	911,280
Spain	902,703
France	862,516
Netherlands	819,539
Canada	778,508

Derived Fields (Recommended)

Derived Field	Formula	Unit	Description
Unit value (price)	`(Value_1000USD × 1000) / Quantity_tonnes`	USD/tonne	Implied price per tonne — proxy for coffee quality/grade
Flow direction label	`Exporter_Name + " → " + Importer_Name`	String	Human-readable bilateral pair label
Trade balance	Sum exports minus sum imports per country-year	1000 USD	Net trade position; positive = net exporter

Key Differences vs. FAOSTAT TCL Dataset

Dimension	FAOSTAT TCL (`coffee_green_trade_FAOSTAT.csv`)	BACI (`coffee_bilateral_trade_BACI.csv`)
Trade structure	Single-country totals	Bilateral (exporter → importer pairs)
Year range	1961–2024	1995–2024
Items	5 coffee types	2 unroasted coffee codes only
Countries	279 (incl. regional aggregates)	229 exporters / 233 importers (countries only)
Bilateral routing	✗ Not available	✓ Full origin-destination pairs
Data quality flags	✓ FAO flags (A/E/X/I)	✗ Not included (harmonized by CEPII)
Source methodology	FAO national reporting	UN Comtrade, reconciled by CEPII

Usage Notes

- **HS_Code as string:** Always read `HS_Code` as text (`dtype=str` in pandas). Reading as integer strips the leading zero, turning `090111` into `90111`.
- **Nominal USD only:** Values are in current USD. For multi-year comparisons, apply a USD deflator (e.g., US CPI or World Bank MUV index).
- **Zero flows omitted:** BACI only records strictly positive trade flows. The absence of a row for a given exporter-importer-year combination means no trade was recorded, not that it was zero.
- **Quantity nulls:** ~0.7% of rows have missing `Quantity_tonnes`. Exclude these rows when computing unit values or quantity-based aggregations.
- **Re-export hubs:** Countries like Germany, Switzerland, Belgium, and the Netherlands appear as significant exporters despite producing no coffee — they are re-export/transit hubs. Filter by known producer countries for origin-only analysis.